A soy protein substitute for animal meat proteins can provide global phosphorus reduction and recirculation opportunities

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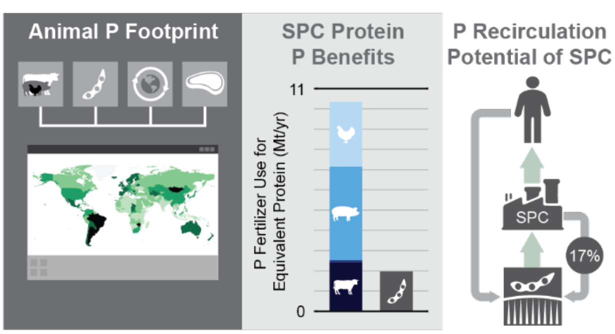
Synopsis: This study analyzes the phosphorus benefits of substitution of animal meat protein with soy protein across 177 countries.

Abstract

Animal meat protein consumption is typically associated with greater environmental impacts than plant-based proteins. While most environmental impact assessments comparing animal and plant-based proteins focus on water usage and carbon emissions, phosphorus (P) use is a key parameter for a sustainable food system. Country-specific animal and crop parameters were leveraged to evaluate the P footprint of current animal meat protein consumption and potential benefits from substitution using soy-based protein. The global average P footprint for animal meat protein consumption when considering synthetic and manure P fertilizers was an estimated 2.6 kg P y^{-1} per capita in 2019. A complete transition to soy protein concentrate (SPC) from animal meat protein was estimated to decrease total P fertilizer applied to animal feed products by 81%, or an estimated 8.3 million metric tons of P per year and 33% of the total estimated 2019 P fertilizer utilization. Additionally, the recovery of P during SPC processing could generate sufficient renewable P fertilizer to replace an estimated 17% of total P utilized in global soybean production. The reduction and efficient reuse of P from a transition to SPC as a primary protein source can generate greater security and resiliency in global food systems.

Graphical Abstract:



Introduction

Phosphorus (P) is an essential crop nutrient critical to global food security.[1] Global P consumption has increased nearly fivefold since 1961 due to mobilization of phosphate rock reserves to support marked increases in crop productivity and human population,[2] but only 22% of this P,[3] and similarly 23% of nitrogen used in food production,[4] is consumed directly by humans in food products. Inefficient P usage can prematurely exhaust currently economically viable phosphate rock reserves,[5] increase volatility in global fertilizer markets,[6] increase wastes associated with P fertilizer production (e.g., phosphogypsum),[7,8] and potentially lead to eutrophication from losses to surface waters.[9] Because food production accounts for nearly 85% of global P utilization,[10] dietary choices directly shape global P demand.

The source of protein in human diets is a key determinant of dietary sustainability. Diets containing a greater proportion of plant-based proteins generally supporting improved health outcomes and lower environmental impacts due to more protective diet scores and potentially lower greenhouse gas emissions, energy demand, and land use.[11] In terms of P footprints, initial estimates indicate that animal-based products have a 8- to 25-times higher P footprint than plant-based products from leguminous crops such as soybeans.[12,13] With global animal-based product demand expected to increase nearly 38%[14] by 2050, a dietary shift towards plant-based proteins could improve diet sustainability and food system resilience.[15,16] However, many desirable high protein crops, including pulses, are not produced at at a rate to substitute animal proteins, meaning that large-scale adoption would require increases in production.[17]

Soy protein foods have a long history as plant-based meat alternatives,[18] and soy production exceeds total pulses production by over four times[19]. Soy protein concentrate (SPC), derived from soybean meal, offers a promising high protein alternative as it is already used in more recent plant-based meat alternatives[20] and provides high digestibility of amino acids with a nearly complete amino acid profile comparable to animal-based proteins.[21] Currently, 98% of global soybean meal is fed to animals as a high protein feed,[22] indicating untapped potential to divert existing SPC for direct human consumption without necessarily intensifying soybean production. However, the potential for SPC to replace animal protein at the global and country level has not been quantified. While land use, water use, and greenhouse gas benefits of dietary plant-based protein alternatives have been generally quantified,[15] the global and country-level P implications of replacing animal protein with SPC remain poorly defined. Moreover, because P can be recovered as a renewable P (rP) fertilizer from wastewater generated during SPC processing,[23] the circularity of P afforded by substitution to soybean protein has not been accounted for in previous studies, underestimating the potential reduction in P footprint.

To fully evaluate the P reduction and reuse benefits of SPC substitution, country-level P footprints and circularity must be individually determined. While previous studies have quantified country-level P footprints for animal meat and plant-based commodities using global average P conversion factors,[12] with few country-specific assessments,[24–28] recent country-level animal and crop data (e.g., feed crop P use efficiency, feed conversion efficiency, animal systems, and international trade) could be leveraged to improve these global estimates. Additionally, the incorporation of trade of animal feedstocks in P footprint estimates can better capture embedded P flows and circularity potential for animal meat and SPC consumption globally.[29]

This study aims to assess the potential of SPC from existing soy production and the associated P benefits of substituting SPC for current animal-based proteins, including potential reductions in dietary P burden and increased P circularity. The specific objectives are to: 1) estimate current P utilization and footprints for animal meat consumption in 177 countries, based on annual synthetic and organic (i.e., manure) P fertilizer inputs, leveraging country- and region-specific parameters and trade data; 2) quantify the country-level potential to substitute SPC, produced from existing soybean meal, for animal meat protein (i.e., beef, goat/mutton, pig, and poultry) and the associated P reduction benefits; and 3) evaluate the circularity potential of rP from SPC processing compared to P recycled via animal manure as a P fertilizer source.

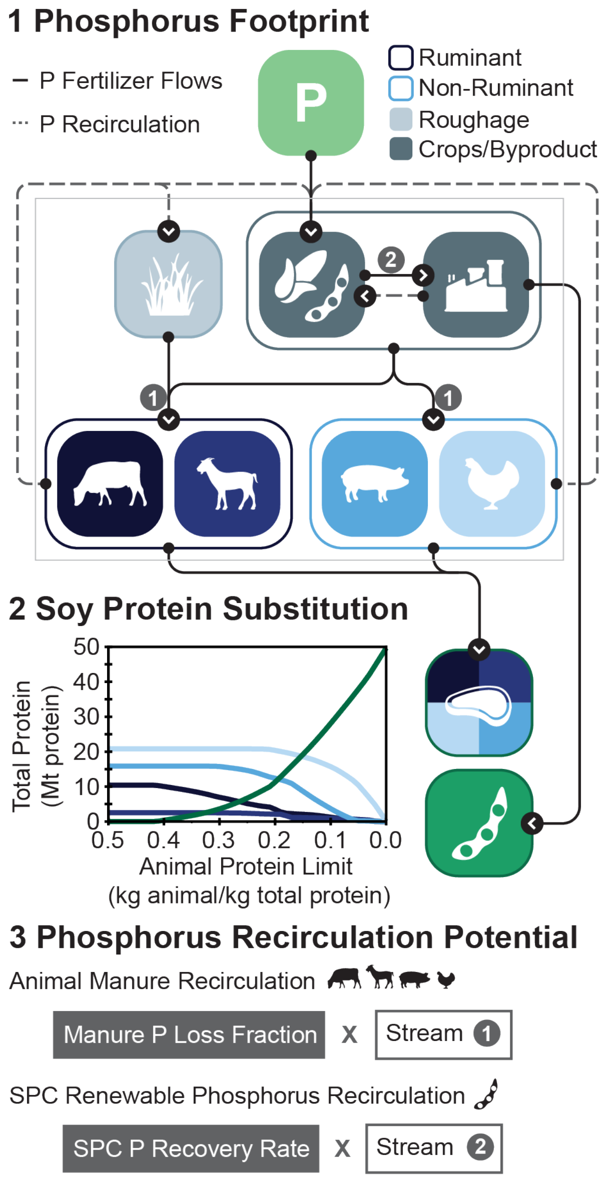


Figure 1. Schematic of P footprint, soy protein substitution, and P recirculation analysis.

Methods

Terminology and Data Sources

The term “P footprint” in this study refers to the total amount of P fertilizer (i.e., synthetic and manure P) utilized to generate a specific animal meat or plant-based protein commodity rather than P losses throughout later stages of food production (**Figure 1**). This is sometimes referred to as the "virtual P" of a commodity in trade analyses, as opposed to the physical P, which is the amount of P physically embedded in the commodity (i.e., P content).[29] All data sources were limited to open access databases for ease of reproducibility. The Food and Agriculture Organization of the United Nations (FAO) statistics database (FAOSTAT),[30] which includes country-level and trade data for primary and processed crops, plant and animal-based food use, and protein consumption was a primary data source for this analysis. Data was limited to 2019 to avoid potential effects of the COVID-19 pandemic. A detailed table of regions used (**Table S1**) and values for parameters (**Table S2-S5**) are provided in the Supplementary Information. Table 1 includes sources and geospatial scale for all parameters used within the methods.

Table 1. Sources for all parameters used in the methods for P footprint, SPC analysis, and P recirculation potential.

Eq #	Parameter	Description	Scale	Source
Phosphorus Footprint Calculation				
1	Q	Food consumption	Country	FAOSTAT[30]
2, 8	FC	Feed conversion efficiency	Regional	Table S2
2, 8	f_{feed}	Feed type fraction	Regional	Mottet et al.[31]
3, 8	C	P content	Global	Tables S3, S4
3	PUE	P use efficiency (by balance)	Country	Zou et al.[32]
Bilateral Trade				
4	M_{net_imp}	Net import quantity	Country	FAOSTAT[30]
4	M_{dsq}	Domestic supply quantity	Country	FAOSTAT[30]
5	M_{net_exp}	Net export quantity	Country	FAOSTAT[30]
5, 6	d	Dry mass fraction	Global	Monfreda et al.[33]
Crop Residues				
6	M	Mass of crop production	Country	FAOSTAT[30]
6	R	Residue to grain ratio	Global	Lal[34]
6	C	P content for residue	Global	Shinde et al.[35]
Phosphorus Recirculation Analysis				
8	P_{loss}	P excreted fraction	Global	Table S5
9	R_{SPC}	P recovery rate	Global	Table S5
9	P_s	Embedded P	Global	Table S5

Animal Meat Phosphorus Footprint

This section outlines the process for calculating the P footprint in each country for consumption of cattle, goat/mutton, swine, and poultry meat. The P footprint represents the total amount of phosphate rock-derived (e.g., synthetic fertilizer) and manure P utilized to produce the feed crops consumed by these animals. Soil residual P was not considered in this analysis but may be utilized to meet crop requirements, which may cause the total estimated fertilizer input to be less than the total embedded P within a crop due to uptake of residual P.[36] Additionally, the recirculation of P in manure was not considered in the P footprint, so manure P recirculated was not subtracted from the overall P inputs and only total P input was considered instead of net P input. The P footprint was calculated for 177 countries that had data available for fertilizer usage and food consumption, allowing for comparison of how dietary changes can impact P needs. The average P footprint for each country was estimated using an approach adapted and modified from Metson et al.[12], shown in Equations (1)-(3). Equation (1) calculates the individual P footprint by country for meat consumption by combining the P contributions from each meat type consumed. Equation (2) calculates a P conversion factor for each animal type by combining the P contributions from all feed types (i.e., roughages, food crops, and byproducts and concentrates) consumed by each animal type. Equation (3) calculates the P inputs for each animal feed type consumed, factoring in both domestically produced and imported feeds.

$$\mathrm{P_{i,t}} = \sum_{\mathrm{k=1}}^{\mathrm{n}} \mathrm{Q_{i,k,t}} \times \mathrm{F_{i,k,t}} \tag{1}$$

$$\mathrm{F_{\,i,k,t}} = \mathrm{FC_{i,k}} \times \sum_{\mathrm{f=1}}^{\mathrm{m}} \mathrm{f_{feed,i,k,f}} \times \mathrm{F_{feed,i,f,t}} \tag{2}$$

$$\mathrm{F_{feed,i,f,t}} = \mathrm{f_{d,i,f,t}} \times \frac{\mathrm{C_f}}{\mathrm{PUE_{i,f,t}}} + \mathrm{f_{nd,i,f,t}} \times \mathrm{F_{feed\_nd,f,t}} \tag{3}$$

where *P* is the individual P footprint in country *i* in year *t*, metric ton (MT) P per capita per year; *Q* is the total meat consumption in country *i* of animal *k* in year *t*, MT meat consumed per capita per year; *F* is the P conversion factor in country *i* for animal *k* in year *t*, MT P required per MT meat; *FC* is the feed conversion efficiency in country *i* of animal *k*, MT total dry matter feed per MT meat;

f_{feed} is the fraction of feed type *f* in country *i* used by animal *k*, MT feed dry matter per MT total dry matter; F_{feed} is the conversion factor between P applied and amount of feed type *f* in country *i* in year *t*, MT P applied per MT feed dry matter; f_d and f_{nd} are the fraction of feed type *f* consumption estimated from domestic production and from imports in country *i* and year *t*, respectively*;* *C* is the P content of feed type *f*, MT P in feed crop per MT feed; *PUE* is the phosphorus use efficiency (PUE) for growing feed type *f* in country *i* in year *t*, MT P in feed per MT P applied; and F_{feed_nd} is the P conversion factor between P applied and amount of non-domestic feed type *f* in year *t*, MT P per MT feed. The primary source and geospatial scale for each parameter can be found in Table 1. In this study, PUE is estimated as the P yield (i.e., P content) over inputs (i.e., P fertilizer and manure), where Zou et al.[32] calculated PUE for each individual crop in each country. This method of PUE calculation is the balance method, which may overestimate PUE relative to the difference[37] or isotopic[19,38] approach. However, because PUE was based on harvested biomass, the P conversion factor for roughage and residue feeds are considered in a separate section, below.

Animal Feed Trade Analysis

Trade plays a pivotal role in distribution of numerous crop-based commodities across the globe. To capture potential benefits to individual P footprints from trade, trade parameters were captured in Equations (4)-(5). Equation (4) calculates the non-domestic fraction of each feed type based on the net imports and the domestic supply quantity of the feed. Equation (5) calculates a weighted average global P conversion factor for each feed type based on countries with net exports to use for non-domestic feed consumption.

$$f_{nd,i,f,t} = \frac{M_{net\_imp,i,f,t}}{M_{dsq,i,f,t}} \tag{4}$$

$$F_{feed\_nd,f,t} = \frac{\sum_{i=1}^{n} M_{net\_exp,i,f,t} \times d_f \times \frac{C_f}{PUE_{i,f,t}}}{\sum_{i=1}^{n} M_{net\_exp,i,f,t} \times d_f} \tag{5}$$

where f_{nd} is the fraction of feed type *f* consumed from imports (i.e., non-domestic) in country *i* in year *t*, MT imported per MT total feed; M_{net_imp} and M_{dsq} is the total net import quantity (i.e., *net_imp = import – export*) and domestic supply quantity (i.e., *dsq = production + import – export*), respectively, in country *i* of feed type *f* in year *t*, MT feed per year; M_{net_exp} is the total net export quantity of feed type *f* from country *i* in year *t*, MT feed per year; and *d* is the dry fraction of feed type *f*, MT feed dry matter per MT feed. The net import and export were used to reduce the effects of intermediary trade countries in which a product is imported and then exported.

Animal Roughage and Residue Feed

Though food crop and co-product/concentrate feeds could be captured by crop categories considered in Zou et al.,[32] there were various roughage and residue feed types from Mottet et al.[31] that could not be directly captured by these categories. However, roughages are a primary feed for grazing ruminants in grasslands, and its inclusion is important to capturing the P utilization for these ruminant animals. To still account for the P embedded in roughage and residues, a PUE of 1.0 was used such that P conversion factors, *F*, were based solely on the roughage and residue P content (i.e., kg P per kg roughage/residue). The P conversion factor was calculated for each roughage/residue similarly to Equation (3) assuming all production was domestic (i.e., $f_d = 1$ and $f_{nd} = 0$). For categories that encompassed multiple crop types (i.e., crop residues and pulse straw), average F factors were generated for each country based on the proportion of each primary crop produced as shown in Equation (6). Since crop residues were used as feed primarily in Asia and Africa[31] (with minor usage for small ruminants in Europe), only the largest contributors to crop residues were considered in those regions (i.e., maize, rice, wheat, and sugarcane).[35]

$$F_{feed,i,residues,t} = \frac{\sum_{k=1}^{n} M_{i,f,t} \times d_f \times R_f \times F_{crop,i,f,t}}{\sum_{k=1}^{n} M_{i,f,t} \times d_f \times R_f} \quad (6)$$

where *M* is mass in country *i* of feed crop *f* produced in year *t*, MT crop per year; *R* is the residue to grain ratio of feed crop *f*, MT residue per MT crop produced; F_{crop} is the conversion factor between P applied and amount in country *i* of feed crop *f* in year *t*, MT P per MT crop.

Soy Protein Analysis

To compare to current animal meat protein consumption, the potential production of SPC from current soybean meal generation was estimated. While soy protein isolate (SPI) is also utilized in plant-based meat products, a majority is SPC. The production of SPC was based on the acid leach procedure in which aqueous ethanol extraction is done. Water is added to soybean meal/defatted soy flour and the pH is adjusted to 4.5 and then centrifuged. The supernatant is removed and the residue is washed with water and centrifuged again producing SPC and a second aqueous supernatant.[23] To determine the total SPC protein potential from soybean meal, or "cake of soya beans" as reported by the FAO, a conversion rate of 20 kg SPC per 47.5 kg soybean meal or 60 kg raw soybeans was assumed.[39] A protein content of 67.4% in SPC (dry-basis)[23] assuming 94% dry matter[40] was used to convert between SPC and protein generation (**Table S5**). All reported "cake of soya beans" from the FAO was also assumed to be capable of being converted into defatted soybean flour.

The substitution of SPC for animal meat protein was evaluated to determine effects on the P utilization and P footprint. Developed countries tend to have the highest animal meat protein consumption.[30] To equitably distribute animal meat reduction goals and to not unfairly burden developing countries with low animal meat consumption limits, substitution was based on limiting the fraction of protein per capita that can come from animal meat within a country (i.e., kg animal protein per kg total protein). This modeling scenario involves high meat-consuming countries reducing their meat consumption prior to low meat-consuming countries. The fraction of animal meat protein consumption was calculated based on FAO estimates of the "protein supply quantity (MT)" for bovine, goat/mutton, pig, and poultry meats in each country divided by the total protein supply quantity of the country as shown in Equation (7).

$$f_{an\_prot,i,t} = \frac{\sum_{k=1}^{q} \mathrm{M}_{\mathrm{prot,i,k,t}}}{\mathrm{M}_{\mathrm{prot,i,tot,t}}} \quad (7)$$

where f_{an_prot} is the fraction of total protein in country *i* in year *t*, MT animal protein per MT total protein consumption; M_{prot} is the amount of protein consumed in country *i* of meat *k* (or total) in year *t*, MT meat. The animal meat protein consumption and total protein, M_{prot}, were both from FAOSTAT.[30]

Limitation on the fraction of animal meat protein was evaluated from 0.5 to 0.0 kg animal meat protein per kg total protein. All animal meat protein in excess of the limit was substituted with SPC protein. The substitution process was done by reducing the amount of animal-based protein by the meat type with the highest estimated P contributions (i.e., 1 bovine meat, 2 goat/mutton meat, 3 pigmeat, and 4 poultry meat). The total P utilization for consumption of each protein type was estimated using Equation (1).

Phosphorus Recirculation Analysis

Phosphorus Potential from Livestock Manure

Manure is the primary source of P recirculation potential in the meat production process. For comparison of P recycling potential between animal meat and soy-based protein production, the amount of P in manure that could be recirculated from the production of bovine, goat/mutton, pig, and poultry meats was determined. The total P available in manure per kg of animal meat consumed in a country was calculated based on Equation (8).

$$\mathrm{P}_{\mathrm{man,i,k}} = \mathrm{P}_{\mathrm{man\_loss,k}} \times \mathrm{FC}_{\mathrm{i,k}} \times \sum_{\mathrm{f=1}}^{\mathrm{m}} \mathrm{f}_{\mathrm{feed,i,k,f}} \times \mathrm{C}_{\mathrm{f}} \quad (8)$$

where P_{man} is the P recirculation potential in manure per meat consumption in country *i* of animal-based commodity *k*, MT manure P per MT meat consumed; P_{man_loss} is the fraction of total P intake excreted in manure for animal type *k*, % or MT P per MT P intake. The fraction of total P intake excreted in manure by animal type, P_{man_loss}, used was 80% for bovine,[41] 48% for goats/mutton,[42] 50% for pigs, and 70% for poultry broilers.[43]

Phosphorus Recovery from Soy Protein Concentrate

Most soybeans are crushed to generate meal and to extract oil for usage in food and biodiesel production. This approach provides centralized locations for soybean processing and potential for P recovery. Laboratory analyses have shown the potential to recover P from further processing of defatted soybean flour (i.e., finely ground soybean meal with oil extracted) into SPC.[23] During production of SPC, aqueous waste streams (i.e., supernatant) are formed from centrifugation that are rich in P, and require further treatment before discharge to the environment at a water resource recovery facility. The P in these waste streams can be precipitated as calcium phytate similarly to rP generation in corn ethanol biorefineries.[23] A general equation for rP potential from SPC production was generated based on the amount of soybean meal processed in Equation (8).[23]

$$\mathrm{rP_{SPC,i}} = \mathrm{R_{SPC}} \times \mathrm{P_{s,i}} \tag{9}$$

where rP_{SPC} is the annual production potential of rP from SPC production in country *i*, MT rP per year; R_{SPC} is the recovery rate of P when generating SPC, %; and P_s is the total embedded phosphorus in soybean meal processed to SPC in country *i*, MT P per year. A baseline of 29.5% recovery of P (R_{SPC}) was used.[23] The uncertainty ranges for the SPC recovery rate were considered in Table S5.

Embedded Phosphorus in Commodities

Analyzing the embedded P in both animal meat and SPC products can help elucidate the potential upstream losses that occur from total P utilization as well as the P burden to water resource recovery facilities. To determine the embedded P in animal meats, standard concentrations of P based on live animal weight were used. The concentrations used for bovine, goat/mutton, pig, and poultry meat were 6.6, 5.4, 5.0, and 5.8 kg P per MT meat, respectively.[44] For SPC, a concentration of 7.6 kg P per MT SPC was used.[45] To ensure that the embedded P was not in excess of that remaining in the feed intake minus that in manure estimates, the calculated embedded P was compared and was limited to that remaining when country and animal level manure estimates were subtracted from the total feed intake (i.e., Remaining P = Total Feed Intake P – Manure P).

Processing Phosphorus Loss Potential

While P can either be embedded in the protein product or captured for reuse in animal wastes and as rP from SPC processing, P is also lost during processing and not recoverable. The potential processing P loss was calculated using Equation (9), below.

$$P_{loss,i,k,t} = P_{i,k,t} - Q_{i,k,t} \times (P_{emb,k} + P_{man,i,k}\ [\text{or } rP_{i,SPC}]) \tag{10}$$

where P_{loss} is the potential processing P loss footprint in country *i* for commodity *k* in year *t*, MT processing P loss per capita per year; $P_{i,k,t}$ is the P footprint in country *i* for commodity *k* in year *t*, MT P per capita per year; P_{emb} is the embedded P in the protein product in protein product *k*, MT P embedded per MT meat or SPC.

Uncertainty and Sensitivity Analysis

Uncertainty in some parameters was evaluated based on how it may affect the global P utilization for animal meat protein consumed and SPC potential, recirculation potential from manure and rP, and the embedded P in protein products. Specifically, uncertainty in the P content of animal feeds and food crops (**Tables S4 & S5**) and the feed P conversion efficiencies for animal protein were evaluated. A total of 10,000 Monte Carlo simulations over the 177 countries were used to determine spearman's rank coefficients of each uncertain parameter. Spearman's rank coefficients are included to demonstrate how sensitive results are to each parameter (**Figure S1**).

Results and Discussion

Animal Meat and Soy Phosphorus Utilization

The total global P utilization of fertilizers (i.e., synthetic and manure P) to produce animal meat protein has increased an estimated 15% from 2014 to 2021, with an estimated 19.8 million metric tons (Mt) P used in 2019 (**Figure 2a**). The P utilization for poultry meat alone increased nearly 37%, particularly in Asia and Africa, while bovine meat increased an estimated 17%. Brazil, China, and the US utilized approximately 44% of all P associated with animal meat protein in 2019. The P associated with bovine meat accounted for an estimated 55 to 58% of total global P use for animal meat protein production, of which 78% was from embedded P in roughage feeds. When considering only non-roughage feeds (i.e., food crops, co-products and concentrates), the total P utilization for animal meat was an estimated 10.5 Mt P per year in 2019, 40 to 48% of which was attributed to pigmeat feeds. Soybean meal was the largest estimated non-roughage contributor to bovine and poultry P use **(Table S6**).

The 2019 global average P footprint of fertilizer use for animal meat consumption was an estimated 2.6 kg P per capita. The global P footprint of bovine meat was an estimated two to three times larger than pig and poultry meat in 2019. Bovine meat was the largest estimated contributor to P use in 80% of countries analyzed due to higher P conversion factors than other meats. Country-level P footprint estimates are presented in Table S8. Countries in Sub-Saharan Africa (SSA) had the lowest P footprints based on their country-level meat consumption, which was lowest for Democratic Republic of the Congo and Mozambique at 0.1 and 0.3 kg P per capita per year, respectively. These low P footprints were due to limited meat consumption, despite high P conversion factors. The largest estimated P footprints from meat consumption were primarily in Latin America and the Caribbean (LAC) at 10.0 (Brazil) and 8.6 (Argentina) kg P per capita per year (**Figure 1b**). These large footprints were primarily due to high meat consumption and P conversion factors (**Figure S2**). A high P conversion factor was the result of poor feed conversion efficiencies and elevated feed P usage due to below-average crop PUE, likely reflecting P-fixation and thus low fertilizer PUE in the generally weathered soils in Brazil. In particular, the LAC region ruminant animals consume a larger amount of fresh grass as part of their diet, which often leads to lower feed conversion efficiencies due to greater consumption to meet nutrition and caloric requirements.[31] Though grasslands are often seen as self-sustaining due to local cycling of manure nutrients, the lack of supplemental P fertilizer and the transfer of manure to croplands has led to an average negative P budget among global grasslands at -1.5 kg P per hectare.[46]

Above average feed crop PUEs and more efficient feed conversion for animal meat proteins can greatly influence country-level animal meat protein P utilization estimates. The United States (US) had the highest meat consumption per capita of all countries in 2019, but its P footprint was estimated to be only 3.3 kg P per capita per year due to more efficient P conversion factors, consistent with a US-specific analysis that estimated 3.4 kg P per capita per year from meat[27]. The use of high-energy feeds in the US, which have higher feed crop PUEs than grazing, contributed to a lower P conversion factor and thus footprint. The consideration of country-specific parameters, such as PUE and trade, had a large influence on individual P footprints: the US P footprint for meat consumption was 23% lower than previous estimates that used a global average P conversion factor.[12] Additionally, the incorporation of trade dynamics accounts for the improved P productivity and complexity associated with feed trade,[47] where maintaining optimal trading is a key component of global P cycling and sustainability.[48] The P footprint results were also compared to country-specific studies, the approach in this study generally yielded higher P footprint results due to the inclusion of manure P inputs, higher per capita meat consumption in 2019, and a resource-based method rather than emissions-based (**Table S7**).

A global sensitivity analysis was conducted to understand key drivers of global P utilization for different animal meats. Sensitivity results generally revealed that ruminant animal and poultry P utilization were most sensitive to changes in P content of feed, whereas pig P utilization had greater sensitivity to the feed conversion efficiency, particularly in East and Southeast Asia (ESEA) (**Figure S3**). The sensitivity of ruminant animals to feed P content is likely due to relatively large consumption of roughages, where roughage P contributions are based on an assumed PUE of 1.0. All animal types exhibited moderate sensitivity to soybean meal P content as the dominant feed type.

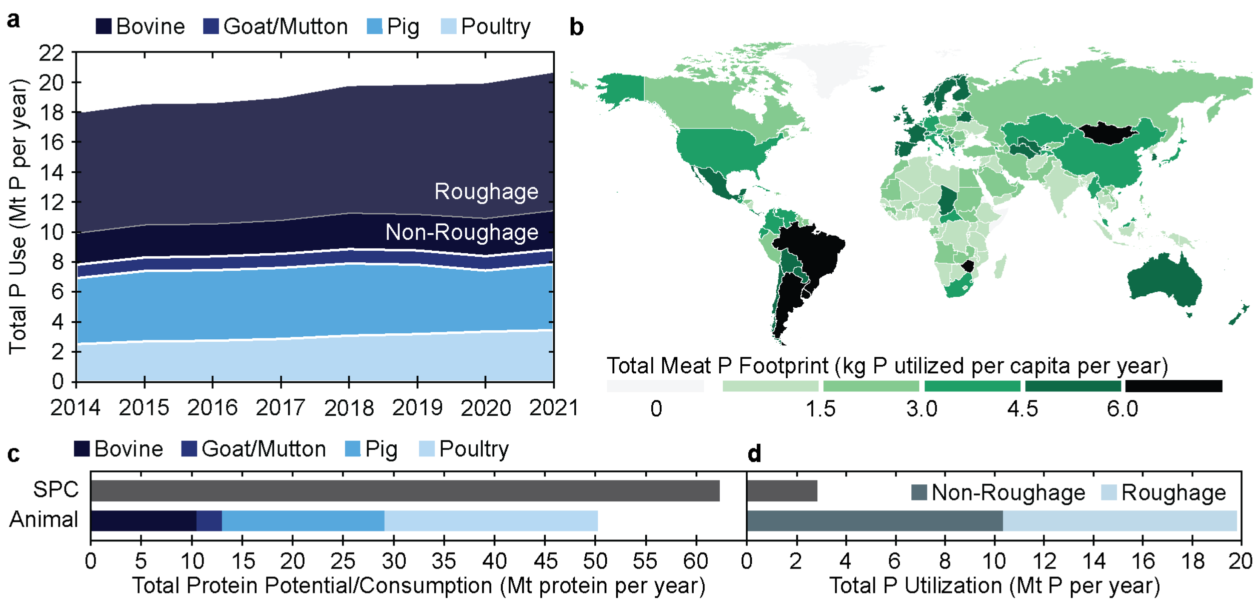


Figure 2. Global phosphorus (P) footprint of meat consumption. (a) Historic global P utilization estimates for production of meat consumed, including the contribution from roughages for bovine meat. (b) Country-level total P footprint estimates (kg P utilized per capita per year) for 2019 bovine, goat, pig, and poultry meat consumption. (c) Estimates of global protein potential for soy protein concentrate (SPC) production from soybean meal and actual 2019 protein consumption of bovine, goat/mutton, pig, and poultry meats. (d) Estimated total P utilization to produce SPC and animal-based total protein, including fraction of P from animal roughage feed.

Soy Protein Substitution Analysis

A transition from animal to plant-based proteins could dramatically reduce global P consumption. The global average P utilization to produce a unit of animal-based protein was an estimated 69 to 96% greater than what is required to generate an equivalent amount of SPC protein from soybean meal. The potential production of SPC from current soybean meal generation, estimated to be 62.3 Mt in 2019, could displace all protein consumption from animal-based meats (50.2 Mt in 2019, **Figure 2c**). The conversion of all soybean meal into SPC rather than animal feed would utilize an estimated 2.8 Mt P per year, approximately 14% of the total P utilized to produce animal-based meat (**Figure 2d**). When considering only non-roughages, production of SPC protein would still only require an estimated 30% of total P utilization of animal-based protein.

Substitution of animal-based proteins with SPC could reduce non-roughage feed P utilization by as much as 81%, or 33% of all P inputs in 2019. The largest estimated P utilization decreases were in the US, China, and Brazil. As complete substitution is unlikely, replacement was evaluated by limiting country-level fractions of total protein consumption from animal sources (i.e., animal meat protein fraction) and substituting meat protein with SPC. When limiting the fraction of protein consumption from animals, substitution would start in the high meat consuming regions of NA, LAC, and Oceania (OCE) followed by WE, Eastern Europe (EE), and ESEA and finally North East and North Africa (NENA), SA, and SSA (**Figure S4**). The hypothetical substitution of animal meat protein with SPC was done by reducing meat protein in succession based on highest overall P conversion factors (i.e., 1 bovine meat, 2 goat/mutton meat, 3 pigmeat, and 4 poultry meat). However, it is likely that the global structure and system of livestock production will change in the future due to climate change and demographic shifts, which could impact the priority reductions. With limited information on dietary and livestock trends, data was restricted to 2019. The change in P footprint for each animal meat by country based on SPC substitution is shown in Table S8.

Human consumption of SPC from country-level soybean meal production could reduce global P utilization by an estimated 12.7 Mt P per year, or a 64% reduction. Comparatively, a 50% substitution of animal products with plant-based meat and milk alternatives by 2050 could nearly halt the net reduction in forest and natural lands and decrease agriculture and land use greenhouse gas emissions by 31%.[15] Substitution with SPC resulted in a consistent global decline in total P utilization across animal-based protein limits, with a more rapid decline in roughage-based P due to initial replacement of ruminant meats (**Figure 3a**). The bovine meat P footprint had the largest country-level reductions up until an animal meat protein fraction of 0.2 (i.e., kg animal to kg total protein), at which point P footprint reductions occurred across all meat types. P utilization for non-roughage feed types (e.g., crops, co-product residues) exhibited a steeper decline starting at an animal meat protein limit of 0.2, with a moderate decline between 0.5 and 0.2. This trend was due to countries with lower animal meat protein consumption favoring monogastric consumption, which primarily consume non-roughages, over ruminant consumption. Additionally, there was only a minimal decline between 0.5 and 0.4 due to the small number of countries that consume greater than 0.4 kg animal protein per kg total protein.

Developed countries can benefit the most from SPC replacement due to their high meat consumption and associated P utilization. However, the P burden from developing countries with increasing meat consumption may outstrip these benefits, particularly considering developing countries had less efficient P conversion factors for animal meat proteins leading to higher P utilization. The largest overall P benefits could be realized by replacing ruminant proteins due to high ruminant P intakes from roughage feeds and relatively inefficient feed conversion. However, when roughage feeds are not considered, monogastric animals had the highest P utilization due to their consumption of feed crops and co-products that are the most highly P fertilized. Replacement of monogastric with plant-based protein could lead to greater decreases in synthetic P fertilizer usage, as has been found for nitrogen (N)[49]. Soybean meal was estimated to be the largest contributor to P utilization of non-roughages, where direct consumption as SPC could avoid the inefficiencies of protein production associated with animal production.

As most soybean meal is fed to animals within the country of cultivation, the current global distribution of soybean meal production could be used to substitute an estimated 73% of total local animal meat protein consumption locally with SPC. The most populous countries including China, India, and the US produce sufficient soybean meal to potentially substitute all animal meat protein consumption (20.5 Mt protein per year) with soy-based protein (**Figure 3b**), reducing the P utilization in these countries by 6.3 Mt P per year of both synthetic and organic fertilizer. To fully substitute SPC for animal-based proteins in countries deficient in soybean meal, an estimated 12.5 Mt SPC protein per year would need to be redistributed from countries where most soybean meal is produced, resulting in an additional 5 Mt P per year reduction (**Figure 3c**).

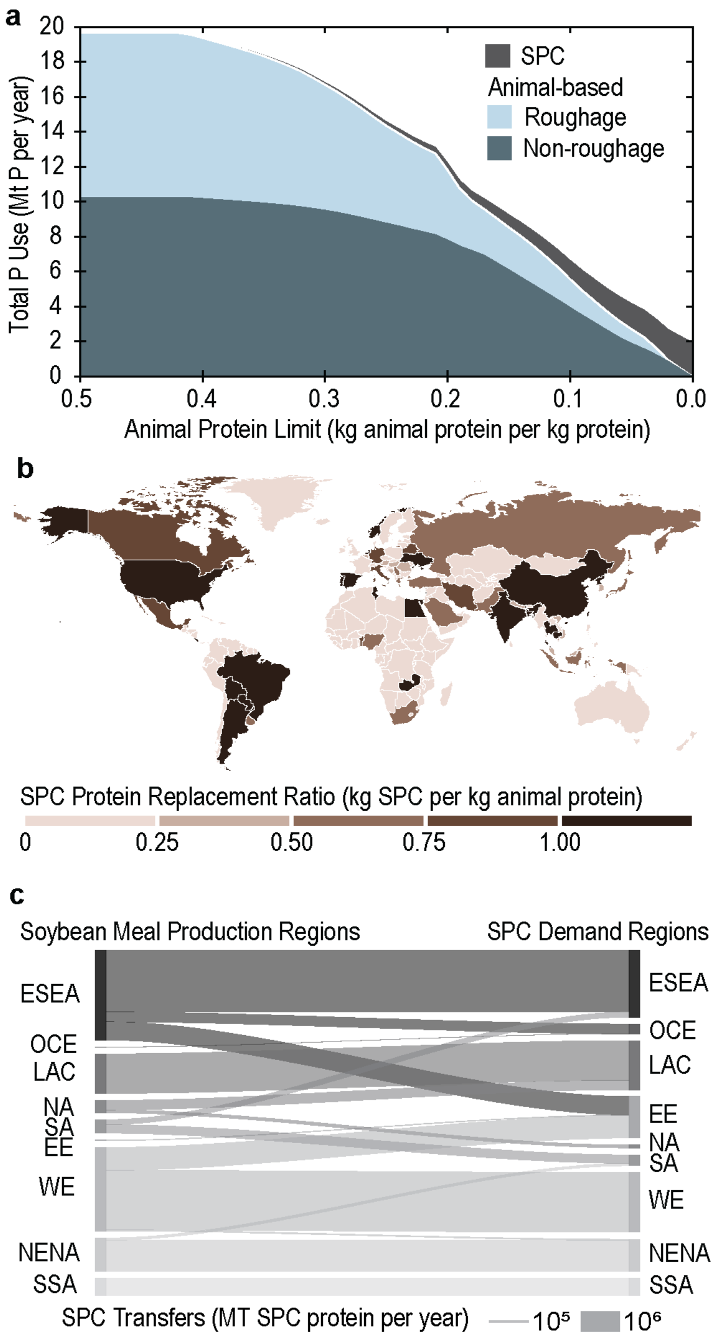


Figure 3. Soy protein substitution analysis. (a) Estimates of the 2019 P utilization for modeled SPC and animal-based protein consumption under varying limits of animal meat protein consumption for each country considering replacement in order of highest/priority P conversion factor (i.e., bovine meat, goat/mutton meat, pigmeat, and poultry meat). (b) Estimates of the ratio between current SPC protein potential from domestically produced soybean meal and animal-based protein consumption. (c) Estimated transfers of SPC protein necessary between major regions to fully substitute animal-based protein

consumption. Regions include East and Southeast Asia (ESEA), Oceania (OCE), Latin America and the Caribbean (LAC), North America (NA), South Asia (SA), Eastern Europe (EE), Western Europe (WE), Near East and North Africa (NENA), and Sub-Saharan Africa (SSA).

Phosphorus Recirculation Analysis

The P footprint of animal meat can be reduced through the reuse of manure nutrients to grow feed. When considering the recirculation potential of P during animal meat production, an estimated 61% (12 Mt P per year) of total global P utilization for animal feeds was embedded in animal wastes that can be recirculated. During SPC processing, rP can also be captured and reused as a renewable P fertilizer in soybean cultivation. Recovery of rP during SPC processing through chemical separation and reuse as a renewable fertilizer could replace an estimated 17% of P fertilizer (0.34 Mt rP per year) used in soybean production globally (**Figure 4a**), with the potential for higher replacement in more agriculturally dense regions such as the Corn Belt of the US. Unlike manure, the production of rP from SPC processing can be more easily increased through process design and phytase incorporation, which can further enhance its circularity potential.[50,51] While manure produced during animal meat production has greater P reuse potential, manure is not a balanced fertilizer that can be easily applied based on specific crop needs due to inconsistent nutrient composition between animals and an N:P ratio that may not match crop needs.[52] Animal manures generally have lower N:P than crop N:P needs, meaning that application of manure to meet crop N requirements will lead to overapplication of P[53] and increase P loss risk to surface waters.[54] Additional barriers including cost, land availability, and transport logistics often lead to manure being treated as a waste material rather than a useful fertilizer and not being strategically applied based on nutrient needs.[55] Alternatively, applying manure on a P basis for crop production would maximize P utilization, and for non-legume crops would co-offset a substantial portion of N fertilizer needs, the remainder of which could be met by supplementary N fertilization. This pollution burden is also seen in the larger greywater requirements of animal products to assimilate added pollution than oil crops such as soybeans.[56] Whereas the P recovered during SPC processing can be applied more efficiently based on crop requirements and easily exported to nutrient constrained areas.

While P can be recovered in animal wastes and SPC as rP, a large fraction of P is embedded in the protein products. The embedded P in animal meat protein and SPC constitutes an estimated 9% (1.8 Mt P per year) and 28% (0.5 Mt P per year) of their total P utilization, respectively. This P is eventually either excreted in waste streams or absorbed by the consumer. Some evidence has suggested that plant-based diets could elevate the P loading to wastewater infrastructure, due to the transition to more P-dense beans and legumes as a protein source.[57] However, one potential benefit is that P wastes become further centralized and can be further recovered at wastewater facilities, rather than lost during other processing steps where it cannot be recovered.

The amount of P lost in processing that is neither embedded in the protein product nor recovered from waste was estimated to be five times higher for animal meat protein (5.9 Mt P per year) than for SPC (1.1 Mt P per year) assuming usage of all potential recycled P (i.e., animal waste or rP). On a country-level, the per-capita processing P loss potential was on average an estimated 5.8 times higher for animal than soy-based protein consumption. Countries in the LAC, WE, EE, and ESEA regions in particular had high processing P losses (up to 3.7 kg P per capita per year) from animal meat protein consumption even when assuming complete recirculation of waste P (**Figure 4b**) due to high P utilization for animal feed crops in those regions from below-average PUEs. The consumption of SPC protein reduced processing P loss potential across most countries, with Brazil having the highest loss potential at an estimated 0.8 kg P per capita per year (**Figure 4c**). The lower processing P loss from SPC substitution is primarily due to reduced P utilization in SPC despite a smaller overall P reuse potential, resulting from better PUEs for soy production and a lower P conversion factor. However, some high meat consuming countries, including the US, have relatively low processing P losses and footprints from animal proteins when compared to SPC due to greater feeding efficiencies at concentrated animal feeding operations (CAFOs). However, the continued intensification of these operations has exacerbated geospatial concentration of manure and P loadings and non-point source pollution.[58] The substitution of animal protein with SPC even in more P efficient countries would help to reduce the burden of CAFOs on water bodies, including the overapplication of P from CAFO manures.

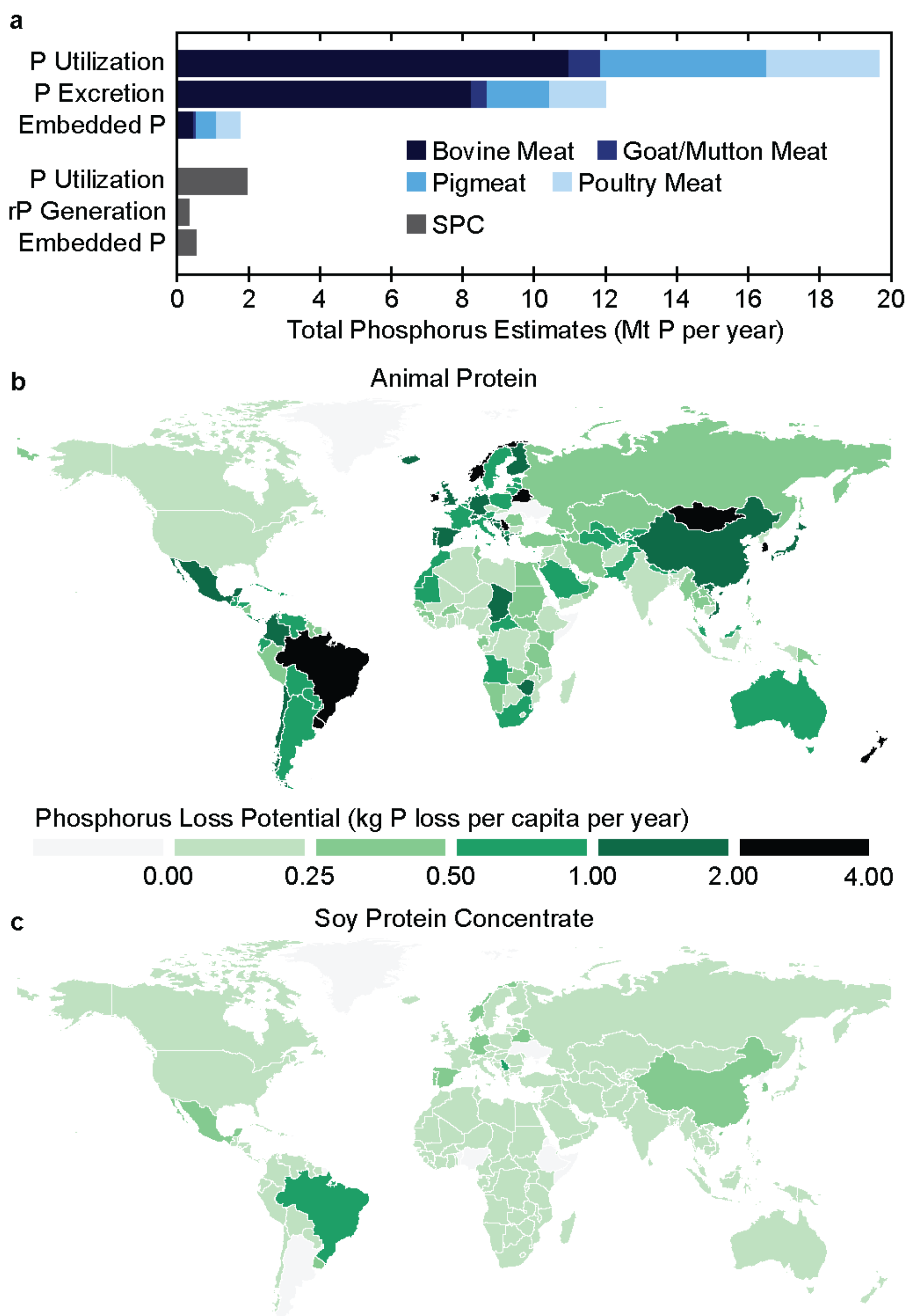


Figure 4. Phosphorus recirculation potential. (a) Estimates of the 2019 total P utilization, P recirculation from animal waste and rP, and embedded P for animal-based protein and SPC, respectively. Estimates of the footprint of processing P loss (i.e., P Utilization – Embedded P – Manure P/rP) for (b) animal meat proteins and (c) SPC.

Study Limitations

This study provides a novel approach to estimating P utilization and savings based on the type of protein consumed at a country-level, accounting for country-specific parameters and trade dynamics into estimates. However, the global scale of this study necessitates several simplifying assumptions were made when generating P estimates. Though many parameters can vary among individual livestock species across varying geographical scales (e.g., feed conversion efficiency, feed ration composition, excretion rates), regional and country-based parameters were used when generating P utilization and recirculation estimates for animal meat protein leading to certain countries appearing more or less P efficient based on the other countries in the region in which they were grouped. Country and intraregional-level values for feed conversion efficiency and feed fraction could improve accuracy of estimates. Another consideration is the change in PUE and livestock composition over time, as this study used 2019 data in its estimates. Though the PUE was based on country-level estimates, PUE often changes as a country develops where it declines during early development, improves during sustainable intensification, and eventually levels off post-development.[32] The methodology for calculating PUE can also influence P estimates, with the balance method used in this study leading likely to overestimates of the total P usage. This leads to limitations in using a fixed year for PUE estimates, as it may improve or decline over time depending on the stage of development of a country. Additionally, the approach used to calculate PUE can be more refined as current approaches often have discrepancies between calculated PUE.[38] The composition of livestock would also change over time, as more countries may intensify livestock production (e.g., industrialized production) which can affect their overall P use.

Outlook

A rapid transition from animal-based proteins is unlikely due to social, cultural, and economic factors,[59] but partial substitution with an SPC alternative protein would provide substantive P benefits by reducing the P intensity of protein consumption. A modest decrease of 0.1 kg animal protein per kg total protein consumed could reduce the national P footprint from meat proteins by more than 50% in at least 107 countries. In countries that eat the most meat per capita like the US, Brazil, and China, the same decrease of 0.1 kg animal protein per kg total protein consumed could reduce animal meat P footprints by an estimated 65%, 58%, and 68%, respectively. Such reductions would alleviate pressure on P fertilizer resources while reducing the eutrophication risks.

However, SPC by itself would not be an adequate direct substitute for animal meat and would need to be combined with other plant-based components to be both palatable[15] and to have a similar nutrition profile.[60] Other plant-based ingredients (e.g., sweet potato, sugar cane) can complement SPC to provide a comparable nutritional profile to animal meats.[15] Added ingredients may also be necessary to overcome social barriers associated with acceptable sensory properties that limit consumer acceptance, including appearance, smell, taste, and texture.[61] However, additional processing of SPC presents trade-offs among nutritional composition, cost, and palatability. Though SPC can offer environmental benefits, the cost and processing needs may still present socio-economic barriers in many countries to local production and use. High-cost alternative proteins may only be possible for wealthier consumers, limiting access to low-income populations. These proteins would also require new processing facilities, supply networks, and markets, where many countries lack the necessary infrastructure and agricultural policy to scale SPC locally. Subsidies and cost reduction potential should be considered and explored to overcome the elevated cost of novel plant-based protein alternatives in comparison to traditional animal meat proteins, as the cost can trigger inequities in consumption potential.[62] Consideration must also be given to potential elevated P concentrations in human excretion from plant-based alternatives,[57] and what benefits and risks could be realized from centralized recovery of the excess P at WRRFs.

This study more broadly addresses the societal challenge of how to manage P sustainability within a dietary transition. While environmental assessments of animal products frequently focus on greenhouse gas emissions and nitrogen losses, P impacts are often understated, including in the Grand Challenges of Engineering identified by the National Academy of Engineering. However, P is a finite resource that is geopolitically concentrated, experiencing increased market volatility, and critical for long-term food security,[63] whereas nitrogen can be generated synthetically and at-scale largely agnostic to geography. Analyzing the P footprint in isolation helps to highlight geographic vulnerabilities in the global food system that is overlooked when considered in aggregate with other nutrients. This study complements multi-impact assessments by demonstrating how soybeans, a widely grown crop, could be leveraged to reduce the P burden in diets. Overall, large P benefits were associated with plant-based SPC protein, where additional P recovery can be incentivized—providing policymakers with additional considerations when generating methods to ensure security and resiliency in food production systems nationally and globally.

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Supplementary Information Available

The Supplementary Information is available free of charge on the publication website.

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